Gramática del analizador sintáctico de Lua 5.5

Martin Moreno Jara

Notación

Los no terminales se escriben así: ⟨NON_TERMINAL⟩ y los terminales en minúsculas y sin llaves triangulares. El símbolo inicial de la gramática es es ⟨CHUNK⟩. ε es la cadena vacía, $\to$ introduce una regla y | separa sus alternativas. La gramática es LL(1).

1. Estructura del programa

Símbolo inicial, bloques y la sentencia `return`.

$$
\begin{aligned}
\langle \text{CHUNK} \rangle &\to \langle \text{BLOCK} \rangle \\
\langle \text{BLOCK} \rangle &\to \langle \text{STATLIST} \rangle \ \langle \text{RETSTAT} \rangle \\
\langle \text{STATLIST} \rangle &\to \langle \text{STAT} \rangle \ \langle \text{STATLIST} \rangle \\
&\mid \varepsilon \\
\langle \text{RETSTAT} \rangle &\to \texttt{return} \ \langle \text{MAYBE_EXPLIST} \rangle \ \langle \text{OPT_SEMILOCON} \rangle \\
&\mid \varepsilon \\
\langle \text{OPT_SEMILOCON} \rangle &\to \texttt{tkn_semicolon} \\
&\mid \varepsilon
\end{aligned}
$$

2. Sentencias

Las asignaciones y las llamadas se factorizan después del primer nombre.

$$
\begin{aligned}
\langle \text{STAT} \rangle &\to \texttt{tkn_semicolon} \\
&\mid \langle \text{NAME} \rangle \ \langle \text{VAR_STAT} \rangle \\
&\mid \langle \text{LABEL} \rangle \\
&\mid \texttt{break} \\
&\mid \texttt{goto} \ \langle \text{NAME} \rangle \\
&\mid \texttt{while} \ \langle \text{EXP} \rangle \ \langle \text{DO_BLOCK_END} \rangle \\
&\mid \texttt{repeat} \ \langle \text{BLOCK} \rangle \ \texttt{until} \ \langle \text{EXP} \rangle \\
&\mid \langle \text{IFSTAT} \rangle \\
&\mid \langle \text{FORSTAT} \rangle \\
&\mid \texttt{function} \ \langle \text{FUNCNAME} \rangle \ \langle \text{FUNCBODY} \rangle \\
&\mid \langle \text{LOCAL_DEC} \rangle \\
&\mid \langle \text{GLOBAL_DEC} \rangle \\
\langle \text{VAR_STAT} \rangle &\to \texttt{tkn_opening_bra} \ \langle \text{EXP} \rangle \ \texttt{tkn_closing_bra} \ \langle \text{VAR_STAT} \rangle \\
&\mid \texttt{tkn_period} \ \langle \text{NAME} \rangle \ \langle \text{VAR_STAT} \rangle \\
&\mid \langle \text{FUNCTIONCALL} \rangle \ \langle \text{FUNCALL_STAT} \rangle \\
&\mid \langle \text{MORE_VAR} \rangle \ \texttt{tkn_assign} \ \langle \text{EXPLIST} \rangle \\
\langle \text{FUNCALL_STAT} \rangle &\to \texttt{tkn_opening_bra} \ \langle \text{EXP} \rangle \ \texttt{tkn_closing_bra} \ \langle \text{VAR_STAT} \rangle
\end{aligned}
$$

$$
\begin{aligned}
&\mid \texttt{tkn_period}\ \langle \text{NAME} \rangle\ \langle \text{VAR_STAT} \rangle \\
&\mid \langle \text{FUNCTIONCALL} \rangle\ \langle \text{FUNCALL_STAT} \rangle \\
&\mid \varepsilon \\
\langle \text{LABEL} \rangle &\rightarrow \texttt{tkn_goto}\ \langle \text{NAME} \rangle\ \texttt{tkn_goto} \\
\langle \text{DO_BLOCK_END} \rangle &\rightarrow \texttt{do}\ \langle \text{BLOCK} \rangle\ \texttt{end} \\
\langle \text{IFSTAT} \rangle &\rightarrow \texttt{if}\ \langle \text{EXP} \rangle\ \texttt{then}\ \langle \text{BLOCK} \rangle\ \langle \text{ELSEIFSTAT} \rangle\ \langle \text{ELSESTAT} \rangle\ \texttt{end} \\
\langle \text{ELSEIFSTAT} \rangle &\rightarrow \texttt{elseif}\ \langle \text{EXP} \rangle\ \texttt{then}\ \langle \text{BLOCK} \rangle\ \langle \text{ELSEIFSTAT} \rangle \\
&\mid \varepsilon \\
\langle \text{ELSESTAT} \rangle &\rightarrow \texttt{else}\ \langle \text{BLOCK} \rangle \\
&\mid \varepsilon \\
\langle \text{FORSTAT} \rangle &\rightarrow \texttt{for}\ \langle \text{NAME} \rangle\ \langle \text{FORTYPE} \rangle\ \langle \text{DO_BLOCK_END} \rangle \\
\langle \text{FORTYPE} \rangle &\rightarrow \langle \text{NUMERICAL_FOR} \rangle \\
&\mid \langle \text{GENERIC_FOR} \rangle \\
\langle \text{NUMERICAL_FOR} \rangle &\rightarrow \texttt{tkn_assign}\ \langle \text{EXP} \rangle\ \texttt{tkn_comma}\ \langle \text{EXP} \rangle\ \langle \text{OPT_EXP} \rangle \\
\langle \text{OPT_EXP} \rangle &\rightarrow \texttt{tkn_comma}\ \langle \text{EXP} \rangle \\
&\mid \varepsilon \\
\langle \text{GENERIC_FOR} \rangle &\rightarrow \langle \text{MORE_NAMES} \rangle\ \texttt{in}\ \langle \text{EXPLIST} \rangle
\end{aligned}
$$

3. Declaraciones

`local` y `global`.

$$
\begin{aligned}
\langle \text{LOCAL_DEC} \rangle &\rightarrow \texttt{local}\ \langle \text{LOCAL_OPTS} \rangle \\
\langle \text{LOCAL_OPTS} \rangle &\rightarrow \langle \text{FUNC_DEC} \rangle \\
&\mid \langle \text{ATTNAME_BRANCH} \rangle \\
\langle \text{ATTNAME_BRANCH} \rangle &\rightarrow \langle \text{ATTNAMELIST} \rangle\ \langle \text{OPT_EXPLIST} \rangle \\
\langle \text{GLOBAL_DEC} \rangle &\rightarrow \texttt{global}\ \langle \text{GLOBAL_OPTS} \rangle \\
\langle \text{GLOBAL_OPTS} \rangle &\rightarrow \langle \text{FUNC_DEC} \rangle \\
&\mid \langle \text{ATTNAMELIST} \rangle\ \langle \text{OPT_EXPLIST} \rangle \\
\langle \text{FUNC_DEC} \rangle &\rightarrow \texttt{function}\ \langle \text{NAME} \rangle\ \langle \text{FUNCBODY} \rangle \\
\langle \text{OPT_EXPLIST} \rangle &\rightarrow \texttt{tkn_assign}\ \langle \text{EXPLIST} \rangle \\
&\mid \varepsilon \\
\langle \text{ATTNAMELIST} \rangle &\rightarrow \langle \text{NAME} \rangle\ \langle \text{ATTRIB} \rangle\ \langle \text{NAME_ATTRIB_LIST} \rangle \\
\langle \text{NAME_ATTRIB_LIST} \rangle &\rightarrow \texttt{tkn_comma}\ \langle \text{NAME} \rangle\ \langle \text{ATTRIB} \rangle\ \langle \text{NAME_ATTRIB_LIST} \rangle \\
&\mid \varepsilon \\
\langle \text{ATTRIB} \rangle &\rightarrow \texttt{tkn_less}\ \langle \text{NAME} \rangle\ \texttt{tkn_greater} \\
&\mid \varepsilon
\end{aligned}
$$

4. Funciones

Nombres de función, cuerpos y listas de parámetros.

$$
\begin{aligned}
\langle \text{FUNCNAME} \rangle &\rightarrow \langle \text{NAME} \rangle\ \langle \text{DOTTED_FUNCNAME} \rangle\ \langle \text{COLON_FUNCNAME} \rangle \\
\langle \text{DOTTED_FUNCNAME} \rangle &\rightarrow \texttt{tkn_period}\ \langle \text{NAME} \rangle\ \langle \text{DOTTED_FUNCNAME} \rangle
\end{aligned}
$$

$$
\begin{aligned}
&\mid \varepsilon \\
\langle \text{COLON_FUNCNAME} \rangle &\rightarrow \texttt{tkn_colon}\ \langle \text{NAME} \rangle \\
&\mid \varepsilon \\
\langle \text{FUNCTIONDEF} \rangle &\rightarrow \texttt{function}\ \langle \text{FUNCBODY} \rangle \\
\langle \text{FUNCBODY} \rangle &\rightarrow \texttt{tkn_opening_par}\ \langle \text{OPT_PARLIST} \rangle\ \texttt{tkn_closing_par}\ \langle \text{BLOCK} \rangle\ \texttt{end} \\
\langle \text{OPT_PARLIST} \rangle &\rightarrow \langle \text{PARLIST} \rangle \\
&\mid \varepsilon \\
\langle \text{PARLIST} \rangle &\rightarrow \langle \text{NAME} \rangle\ \langle \text{PARLIST_TAIL} \rangle \\
&\mid \langle \text{VARPARAGRAM} \rangle \\
\langle \text{PARLIST_TAIL} \rangle &\rightarrow \texttt{tkn_comma}\ \langle \text{PARLIST_AFTER_COMMA} \rangle \\
&\mid \varepsilon \\
\langle \text{PARLIST_AFTER_COMMA} \rangle &\rightarrow \langle \text{NAME} \rangle\ \langle \text{PARLIST_TAIL} \rangle \\
&\mid \langle \text{VARPARAGRAM} \rangle \\
\langle \text{VARPARAGRAM} \rangle &\rightarrow \texttt{tkn_varargs}\ \langle \text{OPT_NAME} \rangle \\
\langle \text{OPT_NAME} \rangle &\rightarrow \langle \text{NAME} \rangle \\
&\mid \varepsilon
\end{aligned}
$$

5. Listas

Listas de variables, de nombres y de expresiones.

$$
\begin{aligned}
\langle \text{MORE_NAMES} \rangle &\rightarrow \texttt{tkn_comma}\ \langle \text{NAME} \rangle\ \langle \text{MORE_NAMES} \rangle \\
&\mid \varepsilon \\
\langle \text{MORE_VAR} \rangle &\rightarrow \texttt{tkn_comma}\ \langle \text{PREFIXEXP} \rangle\ \langle \text{MORE_VAR} \rangle \\
&\mid \varepsilon \\
\langle \text{MAYBE_EXPLIST} \rangle &\rightarrow \langle \text{EXPLIST} \rangle \\
&\mid \varepsilon \\
\langle \text{EXPLIST} \rangle &\rightarrow \langle \text{EXPLIST_ITEM} \rangle\ \langle \text{MORE_EXP} \rangle \\
\langle \text{MORE_EXP} \rangle &\rightarrow \texttt{tkn_comma}\ \langle \text{EXPLIST_ITEM} \rangle\ \langle \text{MORE_EXP} \rangle \\
&\mid \varepsilon \\
\langle \text{EXPLIST_ITEM} \rangle &\rightarrow \langle \text{EXP} \rangle \\
&\mid \langle \text{FUNCTIONDEF} \rangle
\end{aligned}
$$

6. Expresiones

Operandos, expresiones prefijo y llamadas.

$$
\begin{aligned}
\langle \text{EXP} \rangle &\rightarrow \langle \text{OPERAND} \rangle\ \langle \text{EXP_PRIME} \rangle \\
\langle \text{EXP_PRIME} \rangle &\rightarrow \langle \text{ARITH_TAIL} \rangle\ \langle \text{CMP_TAIL} \rangle\ \langle \text{LOGIC_TAIL} \rangle \\
\langle \text{OPERAND} \rangle &\rightarrow \langle \text{NAME} \rangle\ \langle \text{PREFIXEXP_PRIME} \rangle \\
&\mid \langle \text{OPERAND_NOT_ID} \rangle \\
\langle \text{OPERAND_NOT_ID} \rangle &\rightarrow \texttt{nil} \\
&\mid \texttt{false} \\
&\mid \texttt{true} \\
&\mid \texttt{tkn_num}
\end{aligned}
$$

$$
\begin{aligned}
&\mid \texttt{tkn_str} \\
&\mid \langle \text{TABLECONSTRUCTOR} \rangle \\
&\mid \langle \text{UNOP} \rangle \ \langle \text{OPERAND} \rangle \\
&\mid \texttt{tkn_opening_par} \ \langle \text{EXP} \rangle \ \texttt{tkn_closing_par} \\
\langle \text{PREFIXEXP} \rangle &\rightarrow \langle \text{NAME} \rangle \ \langle \text{PREFIXEXP_PRIME} \rangle \\
\langle \text{PREFIXEXP_PRIME} \rangle &\rightarrow \texttt{tkn_opening_bra} \ \langle \text{EXP} \rangle \ \texttt{tkn_closing_bra} \ \langle \text{PREFIXEXP_PRIME} \rangle \\
&\mid \texttt{tkn_period} \ \langle \text{NAME} \rangle \ \langle \text{PREFIXEXP_PRIME} \rangle \\
&\mid \langle \text{FUNCTIONCALL} \rangle \ \langle \text{PREFIXEXP_PRIME} \rangle \\
&\mid \varepsilon \\
\langle \text{FUNCTIONCALL} \rangle &\rightarrow \langle \text{ARGS} \rangle \\
&\mid \texttt{tkn_colon} \ \langle \text{NAME} \rangle \ \langle \text{ARGS} \rangle \\
\langle \text{ARGS} \rangle &\rightarrow \texttt{tkn_opening_par} \ \langle \text{MAYBE_EXPLIST} \rangle \ \texttt{tkn_closing_par}
\end{aligned}
$$

7. Tablas

Constructores de tablas.

$$
\begin{aligned}
\langle \text{TABLECONSTRUCTOR} \rangle &\rightarrow \texttt{tkn_opening_key} \ \langle \text{FIELDLIST} \rangle \ \texttt{tkn_closing_key} \\
\langle \text{FIELDLIST} \rangle &\rightarrow \langle \text{FIELD} \rangle \ \langle \text{FIELD_TAIL} \rangle \\
&\mid \varepsilon \\
\langle \text{FIELD_TAIL} \rangle &\rightarrow \langle \text{FIELDSEP} \rangle \ \langle \text{FIELDLIST} \rangle \\
&\mid \varepsilon \\
\langle \text{FIELD} \rangle &\rightarrow \texttt{tkn_opening_bra} \ \langle \text{EXP} \rangle \ \texttt{tkn_closing_bra} \ \texttt{tkn_assign} \ \langle \text{FIELD_VALUE} \rangle \\
&\mid \langle \text{NAME} \rangle \ \langle \text{FIELD_ID} \rangle \\
&\mid \langle \text{OPERAND_NOT_ID} \rangle \ \langle \text{EXP_PRIME} \rangle \\
&\mid \langle \text{FUNCTIONDEF} \rangle \\
\langle \text{FIELD_ID} \rangle &\rightarrow \texttt{tkn_assign} \ \langle \text{FIELD_VALUE} \rangle \\
&\mid \langle \text{PREFIXEXP_PRIME} \rangle \ \langle \text{EXP_PRIME} \rangle \\
\langle \text{FIELD_VALUE} \rangle &\rightarrow \langle \text{EXP} \rangle \\
&\mid \langle \text{FUNCTIONDEF} \rangle \\
\langle \text{FIELDSEP} \rangle &\rightarrow \texttt{tkn_comma} \\
&\mid \texttt{tkn_semicolon}
\end{aligned}
$$

8. Operadores

Operadores binarios por nivel, operadores unarios y nombres.

$$
\begin{aligned}
\langle \text{LOGIC_TAIL} \rangle &\rightarrow \langle \text{LOGICAL_OP} \rangle \ \langle \text{CMP_EXP} \rangle \ \langle \text{LOGIC_TAIL} \rangle \\
&\mid \varepsilon \\
\langle \text{CMP_EXP} \rangle &\rightarrow \langle \text{ARTH_EXP} \rangle \ \langle \text{CMP_TAIL} \rangle \\
\langle \text{CMP_TAIL} \rangle &\rightarrow \langle \text{RELATIONAL_OP} \rangle \ \langle \text{ARTH_EXP} \rangle \\
&\mid \varepsilon \\
\langle \text{ARTH_EXP} \rangle &\rightarrow \langle \text{OPERAND} \rangle \ \langle \text{ARITH_TAIL} \rangle \\
\langle \text{ARITH_TAIL} \rangle &\rightarrow \langle \text{FLAT_OP} \rangle \ \langle \text{OPERAND} \rangle \ \langle \text{ARITH_TAIL} \rangle \\
&\mid \varepsilon \\
\langle \text{FLAT_OP} \rangle &\rightarrow \langle \text{ARITH_OP} \rangle
\end{aligned}
$$

$$
\begin{aligned}
&\mid \langle \text{BITWISE_OP} \rangle \\
&\mid \langle \text{OTHER_OP} \rangle \\
\langle \text{ARITH_OP} \rangle \rightarrow\ & \texttt{tkn_plus} \\
&\mid \texttt{tkn_minus} \\
&\mid \texttt{tkn_times} \\
&\mid \texttt{tkn_div} \\
&\mid \texttt{tkn_floor_div} \\
&\mid \texttt{tkn_power} \\
&\mid \texttt{tkn_mod} \\
\langle \text{RELATIONAL_OP} \rangle \rightarrow\ & \texttt{tkn_less} \\
&\mid \texttt{tkn_leq} \\
&\mid \texttt{tkn_greater} \\
&\mid \texttt{tkn_geq} \\
&\mid \texttt{tkn_equal} \\
&\mid \texttt{tkn_neq} \\
\langle \text{LOGICAL_OP} \rangle \rightarrow\ & \texttt{and} \\
&\mid \texttt{or} \\
\langle \text{BITWISE_OP} \rangle \rightarrow\ & \texttt{tkn_bit_and} \\
&\mid \texttt{tkn_bitex_or} \\
&\mid \texttt{tkn_bit_or} \\
&\mid \texttt{tkn_right_shift} \\
&\mid \texttt{tkn_left_shift} \\
\langle \text{OTHER_OP} \rangle \rightarrow\ & \texttt{tkn_concat} \\
\langle \text{UNOP} \rangle \rightarrow\ & \texttt{tkn_minus} \\
&\mid \texttt{not} \\
&\mid \texttt{tkn_length} \\
\langle \text{NAME} \rangle \rightarrow\ & \texttt{id} \\
&\mid \texttt{print} \\
&\mid \texttt{pcall} \\
&\mid \texttt{error} \\
&\mid \texttt{warn} \\
&\mid \texttt{xpcall}
\end{aligned}
$$